



Research Note

Identifying influencers on social media

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ABSTRACT

The increased availability of social media big data has created a unique challenge for marketing decision-makers; turning this data into useful information. One of the significant areas of opportunity in digital marketing is influencer marketing, but identifying these influencers from big data sets is a continual challenge. This research illustrates how one type of influencer, the market maven, can be identified using big data. Using a mixed-method combination of both self-report survey data and publicly accessible big data, we gathered 556,150 tweets from 370 active Twitter users. We then proposed and tested a range of social-media-based metrics to identify market mavens. Findings show that market mavens (when compared to non-mavens) have more followers, post more often, have less readable posts, use more uppercase letters, use less distinct words, and use hashtags more often. These metrics are openly available from public Twitter accounts and could integrate into a broad-scale decision support system for marketing and information systems managers. These findings have the potential to improve influencer identification effectiveness and efficiency, and thus improve influencer marketing.

1. Introduction

When Feick and Price first conceptualised the concept of market mavens (an individual that possesses and disseminates an extensive array of general knowledge about all things consumer) in 1987 the world was a vastly different place. Microsoft Windows 2.0 was the new operating system of choice, IBM released the company's first laptop computer (weighing in at around 13 pounds), and the M2 Pocket Phone, the first mobile phone that was small enough to be carried in a (very large) pocket, was introduced. Feick and Price described individuals who would have a large network of friends and co-workers who would diligently read trade magazines and newspapers and generally be people of consumer-based influence in their local areas. Fast-forward more than 30 years. Feick and Price's humble mavens of 1987 would be engulfed by the figurative explosion of social media – suddenly able to access unlimited information at the touch of a button, and access almost half of the world's population via social media (Appel, 2020).

This expansion of the Internet has elevated mavens, and the general social media influencer to a higher than ever importance in the marketing space. It has been reported that over 70 percent of consumers

more likely to purchase a personally recommended product or service (Hubspot, 2019). Most relevant research has focused on market mavens in offline retail markets (e.g. Goldsmith, Flynn, & Goldsmith, 2003). However, the increased prominence of social media in disseminating word-of-mouth (WOM) messages makes it essential to identify and examine market mavens' social media behaviour. Doing so will enable marketing and information systems managers to more easily identify mavens, which is the first step in influencer marketing. This paper seeks to demonstrate how organisations might more efficiently identify these mavens and influential consumers who are likely to be marketing message propagators (Godes & Mayzlin, 2009; Li, Li, & Zhu, 2016; Roelens, Baecke, & Benoit, 2016). Limited previous research has examined the characteristics and behaviors of social media influencers generally. This is despite the growing interest in influencer marketing (Lou & Yuan, 2019a, 2019b). This paper seeks to address that gap.

The primary practical benefit of identifying market mavens is that they can be used to help spread information about product and service innovation ideas, product launches, and marketing campaigns (Chau & Xu, 2012; Li et al., 2016; Uzunoglu & Kip, 2014). For example, Google and Apple regularly test new products with key online consumers, while

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Lego uses chief consumers as a principal source of new product ideas (Schau, Muniz, & Arnould, 2009; van Doorn et al., 2010). Organisations can feed information to such consumers to help the innovation diffusion process. This is common practice with consumers who are purchase club members (e.g., My Best Buy). However, it is not easy to identify market mavens, either offline or online. Simply finding popular or active consumers can be done using simple social media analytic tools (He, Wu, Yan, Akula, & Shen, 2015), however classifying them as market mavens requires a more in-depth approach.

Traditionally, market mavens are identified via self-report attitudinal and behavioural surveys (Goldsmith et al., 2003; Goldsmith, Clark, & Goldsmith, 2006). However, primary research has three significant limitations to social media marketing. First, it is difficult to get a reliable sample of social media users (Wimmer & Dominick, 2013). Second, it is impossible to identify and segment such consumers manually. Third, self-report data is, by its nature, slow to generate and difficult to keep up-to-date, especially for social media data, which is generated in vast real-time quantities (Pournarakis, Sotiropoulos, & Giaglis, 2017). Today, managers may not need to carry out such primary research but, instead, use *big social consumer data* (Kawaf & Istanbuloglu, 2019). Consumers who engage through social media and share their opinions on various online social media platforms create substantial (*volume*), rapidly expanding (*velocity*), and varied data. Data such as textual posts, videos, images, or audio snippets (*variety*) (He et al., 2015). Information systems (IS) researchers have dedicated substantial resources to identifying and validating approaches that could efficiently crunch big social consumer data successfully. For instance, Kumar, Bhaskaran, Mirchandani, and Shah (2013) demonstrate that companies' measurable social media strategy can determine bottom-line return on investments, such that a company's social media platform is a powerful source of value creation and brand relationship development (Souiden, Ladhari, & Chiadmi, 2019).

Overall, big social data offers the potential to identify market mavens through a combination of publicly accessible metrics. However, efficient and effective identification of such powerful consumers is a significant challenge for both marketers and researchers (Aljukhadar, Senecal, & Bériault Poirier, 2019; Boerman, 2020; Ki & Kim, 2019a, 2019b; Lou & Yuan, 2019a, 2019b). Our research contributes to the body of knowledge by detailing how marketers can identify influential consumers from the vast mass of social media users. Thus, the specific aim of this study is to demonstrate how influencers, specifically market mavens, can be validly and reliably identified on social media. A two-stage methodology was used to achieve this aim. In stage one, we identified self-perceived market mavens through an online survey of social media users. In stage two, we compared the social media behaviours of these market 'mavens' and 'non-mavens' by scraping and analysing their recent social media activity. This staged approach allowed us to understand what mavens do on social media that makes them stand out online. Such a marketing information system, which is applicable on social media, has the potential to be an invaluable tool for a decision maker.

2. Theoretical underpinning

2.1. Influencers

Influencer marketing is one of the fastest-growing areas of marketing, and a key area of marketing research (Audrezet, de Kerviler, & Moulard, 2018; Boerman, 2020; Ki & Kim, 2019a, 2019b; Lou & Yuan, 2019a, 2019b; Mediakix, 2019). On social media, influencers share their opinions on products, services, and brands every day. These influencers can be considered opinion leaders who communicate with a sizeable social network of people that follow them (De Veirman, Cauberghe, & Hudders, 2017).

Social media influencers (SMIs) are very attractive to brands. This is because they are seen by consumers as personal, authentic, credible, and down-to-earth sources of information, with the significant benefit of a

large network and thus audience for the brand (De Veirman et al., 2017; Djafarova & Rushworth, 2017; Schouten, Janssen, & Verspaget, 2020). That could be up to 10,000 followers (micro-influencer), up to one million followers (meso-influencer), or over one million followers (macro-influencer). US marketers have reported influencer marketing to be the second most effective promotional strategy (Audrezet et al., 2018). Similarly, 94 percent of marketers who have used influencer marketing campaigns have found them effective (Ahmad, 2018).

2.2. Two-step flow communication theory

We can draw on two-step flow communication theory to explain the nature of influence on social media (Katz & Lazarsfeld, 1955). This theory holds that opinion leaders, or influencers, play a mediating role in the spread of messages from organisations to individuals (Uzunoglu & Misci Kip, 2014). This role involves filtering and decoding messages so that both the content of messages and the style with which they are communicated, suit an influencer's particular follower base (Jiménez-Castillo & Sánchez-Fernández, 2019). In this way, organisations, influencers, and individual consumers constitute a web of social interactions, which is more complicated than a dichotomous sender-receiver model of communication (Uzunoglu & Misci Kip, 2014).

Influence occurs through the premise of observational learning theory, which holds that individual observers use the information shared with them over time to learn and develop beliefs, attitudes, and behaviours, which simplify their decision-making processes (Bandura, 1977; Jiménez-Castillo & Sánchez-Fernández, 2019).

2.3. Communication-persuasion matrix

We can draw on the communication-persuasion matrix to explain the behaviours of influencers on social media at the tactical level (McGuire, 2001). This model posits that persuasive communication, which is a central tenet of influence, relies on five inputs to determine its effectiveness. These are: source, message, channel, receiver, and context (Lou & Yuan, 2019a, 2019b). There is also a set of twelve proposed outputs. These outputs are beyond the scope of the current study, which is limited to how influencers, specifically market mavens, can be validly and reliably identified on social media.

Source relates to the credibility of the influencer, specifically his/her expertise and trustworthiness (Lou & Yuan, 2019a, 2019b). Source credibility is a well-researched aspect of persuasion, of which there are several conceptualisations. In this study, we draw on the seminal work of Hovland, Irving, and Kelley (1953), who break it into expertise and trustworthiness. Expertise refers to the source's competence and knowledge in the subject area (Ki & Kim, 2019a; McCroskey, 1966, 2019b). Trustworthiness refers to the perception that the source is honest and sincere (Giffin, 1967).

The message and channel inputs in the communication-persuasion matrix are also relevant in this study. *Message* relates to the nature and strength of the information used by the influencer (Ki & Kim, 2019a; McGuire, 2001, 2019b), and *channel* refers to the type of media use to communicate the message (Ki & Kim, 2019a; McGuire, 2001, 2019b). The last two inputs, which are receiver characteristics and context, are also important in the persuasion process. They are not relevant in the current study due to their focus on the receiver end of the process.

3. Literature

3.1. Market mavens

A market maven is one specific type of influencer. Market mavens act as hubs of general market information and actively gather and disseminate information to others. They rely on their expertise, perspectives, and judgements on product and market-related information, and have established credibility with large social media audiences, and influence

on their purchasing decisions (Feick & Price, 1987; Ki & Kim, 2019a, 2019b; Lou & Yuan, 2019a, 2019b). Market mavens differ from influencers and other opinion leaders as they disseminate market-related information. In contrast, opinion leaders are knowledgeable about a specific product or product category or within a particular field of influence, such as parenting (Brancaleone & Gountas, 2007). Mavens are knowledgeable about a wide variety of products and services as well as the place, promotion, and pricing aspects of the marketing mix (Brancaleone & Gountas, 2007; Kiani & Laroche, 2019).

Previous research into the behaviours of mavens paints a consistent picture of confident people who enjoy consuming and being a knowledge hub for others. At the personality level, mavens have higher self-esteem, belongingness, and empowerment than non-mavens (Kiani, Laroche, & Paulin, 2016) and are more open to experiences (Kang & Johnson, 2015). In terms of consumption, mavens shop and spend more than the general population (Goldsmith et al., 2003), are more prone to deals and price-savings (Gauri, Harmon-Kizer, & Talukdar, 2016) and are more likely to try new brands (Ruvio & Shaham, 2007). A study of mavens' online referral behaviour found they were four times more likely to refer products or services to other consumers (Walsh & Elsner, 2012). Mavens are also more likely to use consumption and brands to express their self-concept to others (Saenger, Thomas, & Johnson, 2013). However, most such research has focused on market mavens in the offline realm. Comparatively little is known about mavens' behaviour on social media.

3.2. Market mavens on social media

Social media have changed the way people communicate, collaborate, consume, and create (Aral, Dellarocas, & Godes, 2013). Marketers recognise the need to engage with influential consumers through interactive relationships that foster a range of relational benefits, such as trust, satisfaction, commitment and loyalty (Baek, Ahn, & Chio, 2012; Brodie, Ilic, Juric, & Hollebeek, 2013; Hollebeek, Glynn, & Brodie, 2014).

The market maven concept is particularly applicable to social media. A key aspect of market mavens is high sociability (Feick & Price, 1987) and strong social influence (Clark & Goldsmith, 2005). Social media has created a perfect habitat for market mavens. They can use such media to quickly and easily gather and disseminate market information, therefore becoming a hub of expertise (Aljukhadar et al., 2019; Lou & Yuan, 2019a, 2019b). Indeed, a couple of studies have shown mavens can affect others through social media (Aljukhadar et al., 2019; Kiani & Laroche, 2019). This can make or break brands (Chau & Xu, 2012; Uzunoğlu & Kip, 2014). Thus, the ability to engage with influential market mavens is crucial. However, to do this, managers must be able to identify mavens systematically and efficiently (Pagani & Mirabello, 2011; Rishika, Kumar, Janakiraman, & Bezawada, 2013).

3.3. Dimensions of market mavenism

We seek to demonstrate how marketers and marketing researchers can identify broadly influential consumers, specifically market mavens, on social media. We propose market mavens can be identified, and their behaviour better understood using four social media-specific dimensions. Mavens are likely to generate and share information (Abratt, Nel, & Nezer, 1995; Barnes & Pressey, 2012; Feick & Price, 1987) with their substantial network (Laughlin & Macdonald, 2010). Therefore, we included *user network* (i.e., number of followers and listings) and *user behaviour* (i.e., number of posts, percentage of shares, and percentage of interactions) and in the present study. It is vital to test a range of relationships around network and behaviour, where simply having a higher number of posts or followers is not necessarily indicative of influence (Aljukhadar et al., 2019; Djafarova & Rushworth, 2017).

Market mavens must also communicate in a clear way (Clark & Goldsmith, 2005) in order to influence others (Goldsmith et al., 2006;

Ho & Dempsey, 2010; Walsh & Elsner, 2012). Therefore, we also examined *message readability* (i.e., Automated Readability Index [ARI]) and *message structure* (i.e., average percentages of uppercase letters, distinct words, and hashtags). These four dimensions are discussed in subsequent sections.

3.3.1. User network

In line with two-step communication theory, market mavens play a mediating role between brand and consumer, collating, and disseminating relevant product and market-related information (Uzunoğlu & Misci Kip, 2014). In short, mavens often feel obligated to share information to help others (Feick & Price, 1987; Walsh & Elsner, 2012). As a result, they are more likely to disseminate specific knowledge across a range of product categories (Abratt et al., 1995; Barnes & Pressey, 2012) and a wider market scope (Elliott & Warfield, 1993). For example, market mavens often display an earlier awareness of new products and brands compared to other consumers (Clark & Goldsmith, 2005; Ki & Kim, 2019a, 2019b). The dual effects of the market mavens' efforts to share helpful information combined with the need others have for such information mean that mavens are likely to be situated within a substantial network (Laughlin & Macdonald, 2010). This is in line with network theory, which asserts that a maven would be influential if he/she is situated in a dense and central relationship portfolio (Kietzmann, Hermkens, McCarthy, & Silvestre, 2011; Aljukhadar et al., 2019). This is also related to media dependency theory. This theory asserts that a maven's followers will become dependent on them for useful and reliable information that helps with purchase decisions (Ball-Rokeach, 1985; Jiménez-Castillo & Sánchez-Fernández, 2019). Thus:

H1a. *Market mavens have a significantly higher number of social media followers than do non-mavens*

It is possible to classify users in a variety of ways on social media. On Twitter, this is possible through the 'list' function. Twitter users perceived as credible sources of product and market-related information, such as mavens, are likely to more often be added to a list of influencers. Mavens tend to have greater self-efficacy that gives them the confidence to stand out from the crowd and help others (Abratt et al., 1995; Geissler & Edison, 2005). This is particularly true on social media (Aljukhadar et al., 2019; Ki & Kim, 2019a, 2019b). Based on this, network theory suggests that influential information providers will be nodes for other consumers. In effect, people are likely to recognise mavens and identify them as valuable sources of expertise.

Across social media, consumer-to-consumer (C2C) communities have formed so that consumers can assist each other (Jiménez-Castillo & Sánchez-Fernández, 2019). These communities exist because consumers value their peers' opinions and often base their consumption decisions on information provided by competent and influential others; market mavens (Geissler & Edison, 2005; Laughlin & Macdonald, 2010; Price, Feick, & Gunskey, 1995). Drawing on two-step communication theory, one of the ways mavens could be differentiated on social media is by other consumers adding them to lists (Uzunoğlu & Misci Kip, 2014). Further, the communication-persuasion matrix emphasises perceived source credibility as a key input into persuasion (McGuire, 2001). Being listed by others on social media infers a perception of expertise and trustworthiness – the key elements of credibility (Lou & Yuan, 2019a, 2019b). Thus:

H1b. *Market mavens have a significantly higher number of social media listings than do non-mavens*

3.3.2. User behaviour

According to the communication-persuasion matrix, the message is a critical input in the persuasion process (McGuire, 2001). Thus, mavens are likely to generate more messages on social media than regular consumers (Abratt et al., 1995; Barnes & Pressey, 2012; Feick & Price, 1987). Similarly, Scheer and Stern's (1992a, 1992b) influence

framework situates the possession and provision of information as a key component of power and influence – further suggesting that than mavens will share more information than non-mavens in their efforts to influence (Ki & Kim, 2019a, 2019b).

Self-presentation or impression management is another concept that is likely to concern mavens (Trammell & Keshelashvili, 2005). Previous research has found that mavens have an individualistic trait, which means they are more confident and prevalent in general and marketplace discussions, and focused on self-promotion (Barnes & Pressey, 2012). They are also more likely to be interested in learning and engaging with technology (Aljukhadar et al., 2019; Belch, Krentler, & Willis-Flurry, 1995; Geissler & Edison, 2005). On social media, the level of activity on one's account is a key indicator of source credibility. Thus mavens will be likely to actively and regularly post informative content about products and brands (Jin & Muqaddam, 2019; Ki & Kim, 2019a, 2019b). Further, mavens may even become psychologically attached to their accounts, deriving value from the reactions their posts garner (Aljukhadar et al., 2019; Uzunoglu & Misci Kip, 2014). Consequently, mavens' overall use of social media platforms, such as Twitter, is likely to be higher than non-mavens, suggesting:

H2a. *Market mavens have a significantly higher number of posts than do non-mavens*

Social media platforms help information sharing and information generation by providing mavens with access to large amounts of information, and the means to share it (Audrezet et al., 2018). As information brokers, mavens act as a conduit between brands and consumers (Ki & Kim, 2019a, 2019b). They collect, retain, and share information on issues such as product quality, prices, sales, availability, and store personnel characteristics (Brancaleone & Gountas, 2007). This behaviour is in line with two-step communication theory, which emphasises the role of intermediaries in breaking down mass-communication into more micro-communications (Uzunoglu & Misci Kip, 2014). Another element of this theory is interpersonal communication, which is essential to influence others (Weimann, 1994).

Previous research has found mavens seek general market information from a variety of sources (Abratt et al., 1995; Feick & Price, 1987; Goldsmith et al., 2003). On social media, one such source will be the accounts of brands and other influencers, which mavens are likely to follow (Jin & Muqaddam, 2019). Research has shown that the most active accounts on Twitter follow six times more accounts than the average user (Pew Research Center, 2019). Mavens are also likely to share helpful content from these accounts with their followers (Uzunoglu & Misci Kip, 2014). The channel of communication is emphasised within the communication-persuasion matrix (McGuire, 2001). In this case, the interactive nature of the channel is likely to be something that mavens exploit, increasing their access to and provision of information and maximising interaction with others within their network (Ki & Kim, 2019a, 2019b). It follows that:

H2b. *Market mavens have more shares as a proportion of their posts than do non-mavens*

Linked to the previous section, there is likely to be a positive relationship between mavenism and interactions with other users on social media (Uzunoglu & Misci Kip, 2014). Parasocial interaction is common in online influence, where the very nature of social media is to be social (Jin & Muqaddam, 2019). Such interaction is an important driver of perceived trust, which is an element of source credibility in the communication-persuasion matrix (McGuire, 2001). It is also a key part of Scheer and Stern's (1992a, 1992b) influence framework. Beyond informational needs, consumers also have social needs (Jiménez-Castillo & Sánchez-Fernández, 2019). This means that they seek out interactive communications, that is clearly facilitated by social media (Ki & Kim, 2019a, 2019b).

From a maven's perspective, their altruistic behaviours around information sharing tend to be driven by an interpersonal need for

inclusion or belonging (Price et al., 1995). Mavens also tend to be more agreeable than non-mavens (Flynn & Goldsmith, 2017). Consequently, mavens have been found to both initiate discussions about marketplace information and respond to requests from others for such knowledge (Feick & Price, 1987). In this way, mavens can achieve one-to-one social gratification through their information generating or sharing behaviour (Slama & Williams, 1990). Recent research has found that mavens also engage more frequently in online media recommendations than non-mavens (Kiani & Laroche, 2019). They are likely to do this through direct interactions with other social media users. Thus:

H2c. *Market mavens have significantly more direct user interactions as a proportion of their posts than do non-mavens*

3.3.3. Message readability

The nature of the message is a crucial input in the communication-persuasion matrix (McGuire, 2001) and within the influence framework (Scheer & Stern, 1992a, 1992b). Here, where mavens seek to influence their wider social network (Goldsmith et al., 2006; Ho & Dempsey, 2010), the message's style is important (Antioico & Coussement, 2018). The self-presentation concept, which includes ingratiation and competence, also emphasises that mavens should be aware of their communication style (Trammell & Keshelashvili, 2005).

By definition, mavens are acutely aware that other consumers seek them out for information, which is a role they encourage and embrace (Feick & Price, 1987). Indeed, mavens pass on information to others to cement their social position as an information provider (Feick & Price, 1987; Walsh & Elsner, 2012). Thus, they are likely to write in a way that engages other consumers when they are communicating their expertise (Ho & Dempsey, 2010). Stylistic cues, such as stylistic punctuation, exclamation marks, and emoji use, drive perceived influencer authenticity (Audrezet et al., 2018; Jin & Muqaddam, 2019; Ki & Kim, 2019a, 2019b). There is also a need to be clear and objective (Audrezet et al., 2018). Warranting theory explains how perceived authenticity is dependent on clear communication. Recent research has found clear user-generated content (UGC) to have a higher warranting value than subtle advertising messages (Jin & Muqaddam, 2019). Reactance theory also helps explain why consumers do not respond positively to 'forced' persuasion attempts (Jin & Muqaddam, 2019).

In short, mavens seek to draw on the megaphone effect. This effect describes how social media enables individuals to influence groups of others by demonstrating their personal taste in a clear way (Ki & Kim, 2019a, 2019b; McQuarrie, Miller, & Phillips, 2012). The readability of their posts is thus of utmost importance. The Automated Readability Index (ARI) is a popular approach to measuring text readability (Smith & Senter, 1967) that produces an approximate representation of the US school grade level needed to comprehend the text. The specific focus of the ARI is characters per word, rather than the usual syllables per word. This means it is quicker to calculate but still reliable. We expect market mavens' posts to be more readable than non-mavens, due to their desire to engage with the broader consumer group, suggesting:

H3. *Market mavens have significantly more readable posts than do non-mavens*

3.3.4. Message structure

As previously stated, the nature of the message is a key input in the communication-persuasion matrix (McGuire, 2001) and in the influence framework (Scheer & Stern, 1992a, 1992b). Here, we focus on several dimensions of message structure.

Market mavens place an emphasis on self-esteem (i.e. the extent to which a person has a positive attitude about him- or herself; Rosenberg, 1979). In fact, Flynn and Goldsmith (2017) found mavenism to be positively related to the personality trait of extraversion, which enables mavens to distinguish themselves and maintain their social position (Goldsmith et al., 2006; Ho & Dempsey, 2010). Extraversion may also be related to mavens' need to stand out and avoid similarity with others

(Tian, Bearden, & Hunter, 2001). The use of upper case in posts would be one way to attract attention (Brook & Chillis, 2008). Previous work on influencers has proposed that they tend to use more capital letters than non-influencers (Audrezet et al., 2018). Interestingly, there is a generally accepted etiquette online that upper case is akin to shouting, or attention seeking (Brook & Chillis, 2008), and Goldsmith et al. (2003) have found that mavens are more assertive than other consumers. Such behaviour is supported by the self-presentation concept, which includes the role of self-promotion, even intimidation, in influence (Trammell & Keshelashvili, 2005). It follows that:

H4a. *Market mavens have significantly more uppercase letters per post than do non-mavens*

As part of message structure, we are also interested in the percentage of distinct words in mavens' posts. Due to the 'noise' on social media, it is difficult for individuals and brands to make their message stand out, particularly if their messages are divergent in style and content (Audrezet et al., 2018). A maven's posts on social media represent their social media 'brand'. Thus, using consistent language and words will have a positive effect on the level to which consumers engage with their posts on an ongoing basis (Audrezet et al., 2018). According to McGuire's (2001) communication-persuasion matrix and Scheer and Stern's (1992a, 1992b) influence framework, the ability to communicate expertise succinctly is one key to maintaining influence. Particularly relevant is the attractiveness dimension of the influence framework, which implies that the extent to which content is displayed attractively determines its level of influence (Ki & Kim, 2019a, 2019b).

The official Twitter business website notes that posts (tweets) with less than 100 characters achieve 17 percent higher levels of engagement, in terms of shares (retweets), replies and likes (Twitter, 2020a). Studies around online promotion have consistently found increased message length reduces audience attention and engagement (Baltas, 2003). With this prior research in mind, it is likely that mavens keep their posts to fewer distinct words to gain more attention and engagement, suggesting:

H4b. *Market mavens have significantly less distinct words per post than do non-mavens*

Using hashtags facilitates precise information classification and search on social media. Where a maven's role is to curate and share relevant product and market-related information, in a crowded online environment, the use of hashtags is likely to be essential to stand out (Geissler & Edison, 2005; Ki & Kim, 2019a, 2019b). On Twitter, tweets with hashtags achieve 100 percent more engagement (Twitter, 2020b). In line with the influence framework, the use of hashtags could assist with perceived prestige (quality of content), expertise (showcasing knowledge), and attractiveness (Scheer & Stern, 1992a, 1992b). Ki and Kim (2019a, 2019b) specifically found that the attractiveness of social media posts, which includes the use of hashtags, positively affected perceived influence. The self-presentation concept also infers that the use of hashtags could assist with ingratiation and self-promotion, and convey competence (Flynn & Goldsmith, 2017; Trammell & Keshelashvili, 2005).

Looking at the characteristics of mavens, they provide more specific information than other consumers (Abratt et al., 1995), they are comfortable with technology and have high self-efficacy (Geissler & Edison, 2005; Ho & Dempsey, 2010; Ruvio & Shaham, 2007), and they are acutely aware of social media norms (Aljukhadar et al., 2019). Given that the hashtag function is a well-established behavioural norm, it is expected that mavens would be expected to utilise it to reach out to and influence consumers (Clark & Goldsmith, 2005; Erz, Marder, & Osadchaya, 2018).

Moreover, there are laws around the use of influencers and influencer marketing, such as the European Advertising Standards Alliance [EASA] 2018, the Federal Trade Commission [FTC] 2017, and the Word of Mouth Marketing Association [WOMMA] 2017. These require hashtags such as #paidad and #sponsored to be present if a brand is

compensating an influencer (De Veirman & Hudders, 2019). Based on these arguments, we propose that:

H4c. *Market mavens include significantly more hashtags per post than do non-mavens*

4. Methods

4.1. Twitter

Twitter was launched in 2006 and is now a dominant social network, with a growing number of monetizable daily active users. In 2019 Twitter had 145 million, a 17 % increase compared to 2018 (Twitter, 2019). Further growth is forecast in 2020 (Statista, 2019). Over 500 million tweets are posted each day, mostly from Americans, who make up 20 percent of the user base (Twitter, 2019). In the U.S., 80 percent of tweets come from 10 percent of users (Pew Research Center, 2019).

As a brief introduction to the process and nomenclature of Twitter, messages are called tweets, which can contain text, images, videos, links, and hashtags. By default, tweets are publicly available. However, in a practical sense, users are usually only aware of tweets from other users they choose to follow. Thus, each user has a list of followers (users who follow them) and followings (users they follow). Users are identified through Twitter handles (e.g. @Elsevier). Even public tweets can be directed at specific other users, enabling a level of interaction.

In addition to posting original tweets, it is also common to retweeting other users' tweets. For example, if an individual follows @Elsevier and Elsevier tweets about an upcoming IJIM special issue, this user can retweet the message to all of their followers. Retweeting rapidly disseminates messages across user networks, creating a wider sphere of influence. Another relevant element is the list function, which allows users to create separate lists of users they are following. For example, users can create a list of the sports organisations they follow, which will display only these tweets. Similarly, users can create a list of the market mavens they follow, which will display only those tweets. Finally, the hashtag function is used to classify information. Thus, running a search on the hashtag #EPL will display a list of all tweets on this subject (English Premier League) from users being followed and from public users not being followed. Most, if not all, social media platforms use a variation of the hashtag function. Table 1 contains a summary of our maven dimensions and their operationalisation on Twitter.

Although there are dozens of social media tools, we decided to use Twitter for several reasons. First and foremost, Twitter is a particularly inviting environment for market mavens, as the user culture is built on promoting general and specific conversations (Kietzmann et al., 2011). In addition to posting original tweets and conversations, users can also like and retweet others' posts, disseminating messages across user networks and creating a wider sphere of influence. Second, the hashtag function makes Twitter open, searchable, and amenable to the public mining of 'big data' (Li et al., 2016; Risius & Beck, 2015). From a market research perspective, Twitter's Application Programming Interface (API) allows the targeted collection and analysis of large amounts of publicly available free data that is mostly unbiased (Risius & Beck, 2015). Third, Twitter is one of the leading social media outlets marketing managers use to engage with consumers (Pournarakis et al., 2017; Smith, Fischer, & Chen, 2012).

4.2. Participants and procedures

This investigation used a two-stage approach, approved by The University of Western Australia's Human Ethics Research Office, with record number RA/4/1/6203. In the first stage, participants were recruited from the Amazon Mechanical Turk (MTurk) marketplace in May 2013. MTurk, and other similar platforms such as Prolific, have become an increasingly popular source of relatively low-cost, convenient, and reliable data (Hulland & Miller, 2018). Several studies have

Table 1
Market Maven Dimensions and Twitter Terminology.

Market Mavenism Dimension	Social Media Variable	Description	Unit of Analysis
User network	Number of followers	The number of people that agreed to receive your messages (or Tweets) through Twitter. If you add someone to your 'followers', you 'follow' them.	#
	Number of listings	The number of people that added you to their curated group of Twitter users.	#
User behaviour	Number of posts	The number of times the user posts a new Tweet.	#
	Percentage of shares	The number of times the user shares a Tweet from someone else divided by his/her total number of Tweets.	%
	Percentage of interactions	The number of times the user interacts with another Twitter user divided by his/her total number of Tweets.	%
Message readability	Automated Readability Index	The user's average readability measure that gauges the understandability of the Tweet, while approximating the representation of the US Grade level needed to comprehend the text. The formula is given below: $4.71 \left(\frac{\text{characters}}{\text{words}} \right) + 0.5 \left(\frac{\text{words}}{\text{sentences}} \right) - 21.43$	US school grade level
Message structure	Average percentage of uppercase letters	The user's average of the Levenshtein edit distance between the original Tweet and the lower case converted Tweet divided by the total Tweet length. The Levenshtein edit distance represents the number of single characters that need to be capitalized to obtain the original Tweet.	%
	Average percentage of distinct words	The user's average of the sum of distinct words in the Tweets divided by the total Tweet lengths.	%
	Average percentage of hashtags	The user's average of the sum of the hashtags (#) used in the Tweet portfolio.	%

investigated the validity of using MTurk in information systems research. The most in-depth of these (Steelman, Hammer, & Limayem, 2014) compared samples from MTurk, a commercial panel, and undergraduate students. This research reported that the data from US-based MTurk participants were similarly valid and reliable compared to the other samples when examining both the technology acceptance model and the expectancy disconfirmation model (though the non-US MTurk sample performed poorly). Similar findings were reported by Smith, Roster, Golden, and Albaum (2016), with again US-based MTurk and a US-based consumer panel producing similarly reliable data, with the non-US MTurk sample proving problematic and low quality. MTurk has also been validated as a useful way to collect longitudinal data – a difficult task in any context (Daly & Natarajan, 2015). One thing that is made apparent in all of these studies is that the quality of data obtained via MTurk is dependent on the efforts that the researcher goes to in order to protect data integrity. A summary of this was provided in the recent work of Riedel and Mulcahy (2019). In the following section, we outline how we obtained the data and the checks-and-balances we put in place to ensure integrity.

To ensure data quality, we restricted the MTurk task to be only accessible to users registered as US-based, with a task acceptance rate of

greater than 96 %, and who had previously completed more than 100 tasks. Recently Jia, Steelman, and Reich (2017) suggested that several online services allow MTurkers outside of the US to register as if they are inside the US. As an extra layer of precaution, we placed an IP-address marker in the hosted Qualtrics survey that disqualified any respondents with an IP address located outside of the US.

Above and beyond these MTurk platform safeguards, it was also necessary to ensure that the Twitter users we recruited were suitable for the study at hand. Thus, we restricted participation to MTurk users who were active Twitter users. We defined this to the users upfront as a) having a Twitter account that was at least 6 months old at the time, and b) having sent at least 100 tweets or re-tweets. Importantly, prior research has shown that MTurk workers are willing to falsify their answers to screening questions and eligibility criteria to qualify for tasks. This has been demonstrated in terms of falsifying or manipulating political affiliation (Siegel & Navarro, 2019), sexual orientation (Chandler & Paolacci, 2017), as well as general demographics (Sharpe Wessling, Huber, & Netzer, 2017). Therefore, we added a further layer of data integrity – stating that tweets sent in the seven days before, and subsequently during the launch of the first data collection phase in June 2013 would not count towards the 100-tweet requirement. This final safeguard was to avoid users simply sending 100 tweets after reading the requirements and then technically qualifying. This level of data integrity precautions served a dual purpose. First, as we were specifically interested in the potential to identify market mavens on Twitter, it was logical only to include people who had used Twitter in the past at a level that suggested they were not newcomers to the platform. Second, this enabled us to avoid recruiting participants who created a new Twitter account to access this study, as these people would also not be relevant to the study.

Participants completed a short survey in which they provided their Twitter handle with consent for us to follow their Twitter account. Verification of account ownership and further written consent to follow the account was obtained by each participant sending a tweet to the first author with the hashtag #ConsentToFollow (these specific tweets were excluded from subsequent analyses). This is a similar procedure to that of Vander Molen, Kaplan, Choi, and Montoya (2018), who had participants follow a fake user profile to gain access to the social media profiles of their sample. Of the 459 users who completed this initial survey, 36 were removed from the analysis, as they still did not satisfy the Twitter account requirement, leaving 423 usable survey responses.

In the second stage of the study, Twitter's application programming interface (API) was used to scrape each user's Tweets on two data collection dates. First, the API was used to download the basic Twitter profile variables in June 2013. We were able to match and scrape Tweets from 370 respondents, 51 % of whom were female, with an average age of 29.02 (SD = 9.55) years old. These demographics are in line with a recent report from Pew Research Center (2019), which found that 52 % of Twitter users are female, with a significantly younger population compared to the US population. This represented a matching rate of 87 %. The dropout was caused by spelling mistakes of Twitter profiles, and some accounts being set to private. A total of 520,700 tweets were obtained in this way. Due to the dynamic nature of social media platforms, and user behaviours on them, we decided to undertake a smaller second data collection phase five years after the first phase. In the second data collection phase, the Twitter API was used to scrape the basic Twitter profiles and tweets from these 370 respondents from June 2018 until May 2019. Given that some accounts were now disabled or were set to private, we were still able to match 156 respondents of the original respondents. 35,450 additional tweets were scraped in the second phase. This resulted in a total analysed text corpus of 556,150 tweets.

Self-reported market mavenism on Twitter was operationalised using three items adapted for the Twitter context from Feick and Price's (1987) seminal scale. These items were adapted to account for the pan-geography of Twitter. Table 2 shows the three scale items, measured on a 7-point Likert-type scale anchored by strongly disagree (1) and

Table 2

The Market Mavenism Construct items - adapted from Feick and Price (1987).

I like introducing new brands and products to my followers.
I like helping my followers by providing them with information about many kinds of products.
My followers think of me as a good source of information when it comes down to new products or sales

strongly agree (7). These items had a Cronbach's alpha coefficient of 0.91, suggesting the scale was very reliable. A median split on the market mavens construct was used to create a high- and low-level group. The high scoring respondents ($M_{\text{Market Mavenism}} = 5.07$, $M_{\text{Age}} = 29.19$, $\text{Ratio}_{\text{Men/Women}} = 1.26$, $n = 172$) were termed mavens and the low scoring group were termed non-mavens ($M_{\text{MarketMavenism}} = 2.33$, $M_{\text{Age}} = 28.86$, $\text{Ratio}_{\text{Men/Women}} = 0.74$, $n = 198$).

5. Results

The descriptive statistics for the mavenism Twitter dimensions are presented in Table 3. The hypotheses were tested in a series of ANCOVA models (with age and gender as covariates), the results of which are presented in Table 4.

As shown in Table 4, mavens had a significantly higher number of other Twitter users following them compared to non-mavens ($M_{\text{high}} = 802.30$; $M_{\text{low}} = 215.00$; $F = 4.14$; $p = .04$; $\eta_p^2 = .013$). This supports H1a. However, H1b was not supported, as mavens were not listed significantly more than non-mavens ($M_{\text{high}} = 9.29$; $M_{\text{low}} = 5.73$; $F = 1.67$; $p = .20$; $\eta_p^2 = .006$). In support of H2a, mavens generated significantly more posts ($M_{\text{high}} = 1713.20$; $M_{\text{low}} = 1320.60$; $F = 8.46$; $p < .01$; $\eta_p^2 = .025$). However, mavens did not have more personal interactions with other Twitter users ($M_{\text{high}} = 23.73$; $M_{\text{low}} = 20.78$; $F = 2.52$; $p = .11$; $\eta_p^2 = .007$) and they did not share more information (i.e. retweet posts from other Twitter users) ($M_{\text{high}} = 14.60$; $M_{\text{low}} = 14.76$; $F = 0.03$; $p = .87$; $\eta_p^2 = .000$). Thus, H2b and H2c were n and H2c were

Contrary to H3, mavens wrote less readable messages than did non-mavens ($M_{\text{high}} = 8.59$; $M_{\text{low}} = 7.48$; $F = 7.19$; $p < .01$; $\eta_p^2 = .021$). However, H4a was supported, as mavens did use significantly more capital letters ($M_{\text{high}} = 8.52$; $M_{\text{low}} = 7.56$; $F = 12.68$; $p < .01$; $\eta_p^2 = 0.034$). Moreover, non-mavens used a greater variety of words when writing their posts ($M_{\text{high}} = 28.32$ vs $M_{\text{low}} = 30.58$; $F = 4.67$; $p = .03$; $\eta_p^2 = .012$), supporting H4b. In addition, H4c was supported as the use of hashtags was higher for the mavens ($M_{\text{high}} = 29.66$ vs $M_{\text{low}} = 24.69$; $F = 4.98$; $p = .03$; $\eta_p^2 = .010$).

6. Discussion

Consumers' increasing use of social media platforms has forced companies to develop innovative ways to analyse and use big social media data. We show that a mixed-method approach where a survey is

Table 3

Descriptive Statistics for the Overall Sample.

Market Mavenism Dimension	Social Media Variable	Mean	SD
User network	Number of followers	488.02	2557.37
	Number of listings	7.39	22.29
User behaviour	Number of posts	1503.11	1250.16
	% of shares	14.68	14.63
Message readability	% of interactions	22.15	17.90
	Automated readability index (US grade level)	7.99	3.83
Message structure	Mean % of uppercase letters	8.01	2.61
	Mean % of distinct words	29.53	10.47
	Mean % of hashtags	27.00	25.44

Table 4

A Summary of the Hypothesis Testing Results.

Market Mavenism Dimension	Social Media Variable	Market Mavenism		F-value
		Mavens Mean (SD)	Non-mavens Mean (SD)	
User network	Number of followers	802.30 (3667.80)	215.00 (642.00)	4.14*
	Number of listings	9.29 (27.19)	5.73 (16.82)	1.67
User behaviour	Number of posts	1713.20 (1306.80)	1320.60 (1171.80)	8.46**
	Percentage of shares	14.60 (15.04)	14.76 (14.29)	.03
	Percentage of interactions	23.73 (18.75)	20.78 (17.05)	2.52
	Automated	8.59 (4.55)	7.48 (2.97)	7.19**
Message readability	Readability Index – US Grade Level			
Message structure	Average percentage of uppercase letters	8.52 (2.75)	7.56 (2.40)	12.68**
	Average percentage of distinct words	28.32 (10.71)	30.58 (10.17)	4.67*
	Average percentage of hashtags	29.66 (28.04)	24.69 (22.76)	4.98*

Note: * $p < .05$; ** $p < .01$; degrees of freedom (1,366).

employed to measure market mavenism combined with empirical social media data, scraped at high volumes and low cost, is a valuable way to identify mavens. This approach could be extended to other domains of interest, such as opinion leadership and complaint propensity.

Our results suggest that mavens can be classified based on the way they use social media (i.e. their network, behaviour, and message characteristics), as can be seen in Table 5. As we hypothesized, mavens tend to be more central within their networks on social media, and tend to use social media differently to non-mavens. This is in line, first, with the overarching theory around influencers upon which we drew; two-step communication theory. It holds that individuals like mavens are intermediaries that have a distinct role in filtering and disseminating product and market-related information (Uzunoglu & Misci Kip, 2014). It is also in line with the communication-persuasion matrix, which holds that mavens will communicate differently, more persuasively, than

Table 5

Summary of Hypotheses and Main Findings.

Market Mavenism Dimension	Hypothesis	
User network	H1a	Market mavens have a significantly higher number of social media followers compared to non-mavens
	H1b	Market mavens have a significantly higher number of social media listings compared to non-mavens
	H2a	Market mavens have a significantly higher number of posts compared to non-mavens
User behaviour	H2b	Market mavens have more shares as a proportion of their posts than do non-mavens
	H2c	Market mavens have significantly more direct user interactions as a proportion of their posts than do non-mavens
Message readability	H3	Market mavens have significantly more readable posts compared to non-mavens
	H4a	Market mavens have significantly more uppercase letters per post compared to non-mavens
	H4b	Market mavens have significantly less distinct words per post compared to non-mavens
Message structure	H4c	Market mavens have significantly more hashtags per post compared to non-mavens

non-mavens (Lou & Yuan, 2019a, 2019b; McGuire, 2001). Previous research into mavens has investigated these aspects of mavens' behaviour in an offline context, but few studies have examined them on social media. In particular, our findings offer a new direction for identifying these influential consumers on social media, and a support system for marketing decision-makers across the world.

Mavens had more followers on social media than did non-mavens (H1a). This extends previous knowledge about the personal network size of mavens (Laughlin & Macdonald, 2010) from an offline to an online setting. It also confirms suggestions about the size of mavens' online networks (Aljukhadar et al., 2019; Kiani & Laroche, 2019). Besides two-step communication theory, network theory, and media dependency theory support the central role of influencers such as mavens in online networks (Aljukhadar et al., 2019; Jiménez-Castillo & Sánchez-Fernández, 2019).

Unexpectedly, mavens do not tend to be 'listed' more than non-mavens (H1b). This finding is not in line with previous research, which found mavens were seen as credible sources and identified as central actors within their network (Aljukhadar et al., 2019; Feick & Price, 1987). On Twitter, it seems that being listed is not entirely representative of being viewed as a credible source (Abratt et al., 1995; Geissler & Edison, 2005). Where consumers value their peers' opinions and use them as the basis for product and service choices (Geissler & Edison, 2005; Laughlin & Macdonald, 2010), further research is needed to give insight into why savvy Twitter users would not list mavens. One practical explanation could be the under-use of Twitter's list function which was a relatively hidden feature in the platform's interface until 2019 (Perez, 2019). Another could be that social media users choose to follow only influential individuals such as mavens, and thus do not require to list them separately.

We found mavens tended to post more than non-mavens (supporting H2a). This result extends findings from prior research in offline settings that found mavens were more likely to generate information in their chosen channels of communication (Abratt et al., 1995; Barnes & Pressey, 2012; Feick & Price, 1987). It is supported by the communication-persuasion matrix (McGuire, 2001) and the influence framework (Scheer & Stern, 1992a, 1992b), both of which place importance on the information provision role of influencers. Previous research has also found mavens value interpersonal interaction, often responding to requests from others for market information (Kiani & Laroche, 2019; Price et al., 1995; Uzunoglu & Misci Kip, 2014). These behaviours lead to mavens posting more often on Twitter.

However, mavens did not share more than non-mavens (rejecting H2b). This hypothesis was based on prior research that emphasised the role mavens play in seeking out and disseminating information within their networks (Audrezet et al., 2018; Goldsmith et al., 2003; Ki & Kim, 2019a, 2019b). It was also founded on the communication-persuasion matrix, where the importance of information provided through interaction with others is emphasised (Ki & Kim, 2019a, 2019b; McGuire, 2001). It may be that, on social media, mavens tend to be original content creators, rather than sharers of other users' content. This could be supported by Scheer and Stern's (1992a, 1992b) influence framework, where the element of prestige in persuasion could lead mavens to maintain some distance from other users. Further research is needed to clarify this issue.

In the same vein, we found that mavens did not have more direct interactions than non-mavens (rejecting H2c). Aside from satisfying social needs, which mavens and their users do have, we proposed that these interactions could be a key means of building source credibility (McGuire, 2001; Scheer & Stern, 1992a, 1992b). However, it seems mavens do not rely on interactivity to build credibility, or indeed to meet social needs. Whether mavens are taking full advantage of the interactive nature of social media channels for influence and persuasion is a potential area for future research.

We found interesting results for *message readability*, as mavens' messages were *less* readable (reversing H3). We had hypothesised

mavens' readability would be better, as they wish to communicate to and engage with wide groups of consumers (Goldsmith et al., 2006; Ho & Dempsey, 2010; Jin & Muqaddam, 2019). Following our previous arguments, the notion of prestige in persuasion may again play a role (Ki & Kim, 2019a, 2019b). Further, in the self-presentation concept, there are elements of self-promotion and intimidation that could lead to write posts in a more complex style (Trammell & Keshelashvili, 2005). It is noted that there are also elements of ingratiation and competence in the self-presentation concept, which we proposed would lead mavens to write more readable posts. Previous research around mavens offline has examined the importance of social position (Feick & Price, 1987; Walsh & Elsner, 2012). It is possible mavens may try to achieve higher status by communicating their expertise through the complexity of the language in their messages.

Our results around *message structure* also followed this line of argument, as we found mavens tend to use more uppercase letters (H4a) and hashtags (H4c) in their posts. The use of uppercase letters confirmed prior research into extraversion and a general need to attract attention through assertiveness (Goldsmith et al., 2003; Ho & Dempsey, 2010; Flynn & Goldsmith, 2017). It may also denote a sense of prestige and intimidating self-promotion (Ki & Kim, 2019a, 2019b; Trammell & Keshelashvili, 2005). The use of hashtags, as a form of information classification, suggests that mavens consider how to present and communicate information on social media (Abratt et al., 1995; Ki & Kim, 2019a, 2019b). It also suggests that they are comfortable with social media use norms (Aljukhadar et al., 2019; Geissler & Edison, 2005; Ho & Dempsey, 2010). Here, perhaps they do care about the attractiveness of their posts (Ki & Kim, 2019a, 2019b), or perhaps they want their information to be noticed through prestige and expertise (Scheer & Stern, 1992a, 1992b; Trammell & Keshelashvili, 2005). Finally, we found mavens tended to use fewer distinct words in their posts (H4b). This supports our suggestion that mavens write succinct and attractive posts, as this increases the likelihood of obtaining attention and engaging their followers (Audrezet et al., 2018; Ki & Kim, 2019a, 2019b).

6.1. Limitations and future research

Although this study shed light on an important and under-investigated aspect of mavenism, there were several limitations. First, our study included only English-speaking American respondents. Wiedmann and Walsh (2001) found general maven characteristics differ across countries, so it is essential to see whether differences exist in the maven classification drivers across different languages and cultures. Second, we used a self-report survey to identify market mavens. Although all major mavenism studies have used a self-report approach, it would be beneficial to have a secondary identification approach, perhaps through peer-observations on social media. Third, while we examined the way Twitter users wrote their posts, a valuable path for future research would be to investigate whether the meaning and content of the posts is a good predictor of mavenism. For example, it is known mavens focus on elements of the marketing mix (Brancaleone & Gountas, 2007). Thus, it could be useful to categorise people's posts around these product, price, promotion, and place categories to estimate the relative impact these categories have on maven identification. Fourth, we collected data on Twitter. It would add value to the discussion to investigate if these results hold on other social media, such as Instagram. Fifth, this research was focused explicitly on the construct of market mavenism and how mavens might be identified on Twitter. Expanding this investigation to take a more nuanced approach is a logical next step in this research stream. As an example, adding constructs such as the motivation behind social media usage to a model (such as need for affiliation or self-aggrandisement) could extend our understanding of the differences between mavens and non-mavens use of Twitter. It might also add value to experimentally manipulate the context for information provision, for instance answering a general market query versus a very specific information request. This could

further elicit differences in the way mavens compared to non-mavens interact with other Twitter users.

We looked at the identification issue (the ‘who’ question) around market mavens, which is a fundamental first step for marketing decision makers. Future research should look at the ‘how’ question to provide insights into the way managers might engage with mavens in their day-to-day business activities, as [Godes and Mayzlin \(2009\)](#) have done with opinion leaders. This study delivered insights into the variables that might be used to identify mavens (the ‘who should I target?’ question). However, managers struggle with deciding how to engage with consumers ([Brodie et al., 2013](#)). It seems managers may need to ‘entice’ influential consumers or mavens to participate through relevant and valuable content, and ‘validating’ their participation by responding to them ([Smith et al., 2012](#)). Therefore questions such as ‘Which channel(s) should be used to contact mavens?’, ‘When should managers contact market mavens?’ or ‘Which type of message should managers use to convince mavens to engage?’ are important to answer in order to optimize an organization’s overall social media strategy.

6.2. Implications for theory

Influence is underpinned in the two-step communication theory, which tends to focus on the dominant network of influencers ([Katz & Lazarsfeld, 1955](#)). It follows that we found that mavens on social media differed from other social media users. They have more followers and post more often. Influence is similarly underpinned by the communication-persuasion matrix ([McGuire, 2001](#)), which tends to focus on the persuasive communication style of influencers. It follows that we found that mavens write in a more complex manner, and use more uppercase letters, fewer distinct words, and more hashtags. We contribute further by investigating the characteristics and behaviours of mavens on social media, where previous research focused on offline settings. In addition, we provide means to identify mavens on social using publicly available data, where previous research focused on self-report mechanisms.

Our classification system for mavenism on social media is based on network and behavioural metrics. This four-dimensional classification framework (user network, user behaviour, message readability, and message structure) provides a strong base upon which to identify mavens and other influencers on social media that could be adapted to most major social media platforms. This contribution is of particular value because, as the power in marketing moves more into the hands of consumers, it is crucial to identify influential consumers systematically and efficiently ([Li et al., 2016](#)). We recommend applying our framework on other social media, including blogs, social networks, and review sites, so as to understand better the activities mavens undertake on social media ([Go & You, 2016](#)).

From a methodological point of view, our multistage data collection approach offers researchers a sound way to further this field of study. Simply recruiting by referrals or spamming Facebook or Twitter accounts is susceptible to self-selection bias if there are not reasonable incentives. It is also impractical to offer incentives without a micro-transaction platform in place. Thus, combining the MTurk micro-transaction platform with an online survey and subsequent public data scraping provides a valid way to recruit. Eventually, being able to rely on big, social data and removing the need for expensive primary market research, such as surveys, is crucial. We have taken one step towards this goal.

6.3. Implications for practice

Our results have significant implications for marketers who seek to use social media to profile and engage with consumers. Social media facilitates consumer-to-consumer interactions around product referrals, service evaluations, and message co-creation. However, some consumers are more influential than others. Researchers need to be able to

identify such key influencers. Managers in companies such as Google, Lego, Finnair, and Infosys have sought to identify and engage with such key influencers. One of the difficulties surrounding the influencer market is going beyond simply the number of followers. This is particularly important given that followers can be bought, and the numbers manipulated with bots. A reliable maven identification approach, that extends beyond the basic ‘look at me, I’m popular’ metrics, would facilitate this task.

We found it is possible to identify market mavens using ‘big data’ relatively quickly and cheaply. High post and follower counts, use of uppercase and hashtags, and shorter posts are all indicative of mavenism. Social media data are created with every user interaction, and, as noted earlier, these data are publicly available at low cost ([Li et al., 2016](#); [Mostafa, 2013](#); [Risius & Beck, 2015](#)). Big data allows inexpensive access to significant large amounts of information through automated scraping and analysis tools ([Pournarakis et al., 2017](#)). Previously, the identification of these influential consumers was possible only through self-reported surveys that would need to be sent to each consumer identified online. This approach is complicated, time-consuming, expensive, and thus infeasible given the thousands of social media users expressing themselves online as consumers of a given company. In contrast to crunching cheap, quick, and unbiased big social data, the use of only self-reported surveys negates the advantages a big data framework provides.

To identify mavens, we can use metrics around the four suggested mavenism dimensions (user network, user behaviour, message readability, and message structure). These metrics are at a general level and do not require managers to delve into mavens’ written content. This improves reaction times and enables decision support systems to be developed that allow managers to identify and react to mavens on social media, which is a key issue and challenge ([Hennig-Thurau et al., 2010](#)).

There are already some tools that identify influencers on social media, such as Commun.it and Twitonomy. However, the methodology behind these tools is unclear and not underpinned by sound theory and synthesised research knowledge ([He et al., 2015](#)). Our approach opens the ‘black box’ and relates a combination of social media metrics to market mavenism dimensions. We would ideally hope that our findings would be integrated into a user-friendly web-based tool. Or indeed used within one of the existing commercially available black-box tools (with permission). On Twitter, these metrics can be accessed using the API function (<https://dev.twitter.com/>). The metrics around user network and behaviour can be directly retrieved from the API function, while the metrics around message readability and message structure require some basic programming skills.

7. Concluding remarks

The present study makes an important contribution that shows how influential mavens on social media can be identified using freely accessible data obtainable through social media APIs. This is a valuable contribution for those doing both academic and practical research who are planning to combine traditional primary research with big data research. It is clear that decision-makers can use big social data to identify influential consumers. Of course, managers will still need to make decisions about how these consumers can help their campaigns and how they will engage with them. However, the approach proposed here has laid the foundations to make this possible.

CRedit authorship contribution statement

Paul Harrigan: Project administration, Funding acquisition, Conceptualization, Writing - original draft, Writing - review & editing. **Timothy M. Daly:** Funding acquisition, Investigation, Methodology, Writing - original draft, Writing - review & editing. **Kristof Coussement:** Methodology, Formal analysis, Investigation. **Julie A. Lee:** Funding acquisition, Conceptualization, Supervision. **Geoffrey N.**

Soutar: Funding acquisition, Conceptualization, Formal analysis, Supervision. **Uwana Evers:** Conceptualization, Data curation, Writing - review & editing.

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